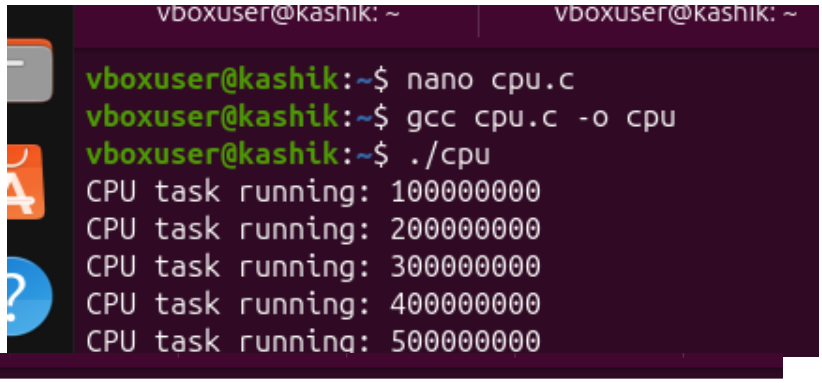
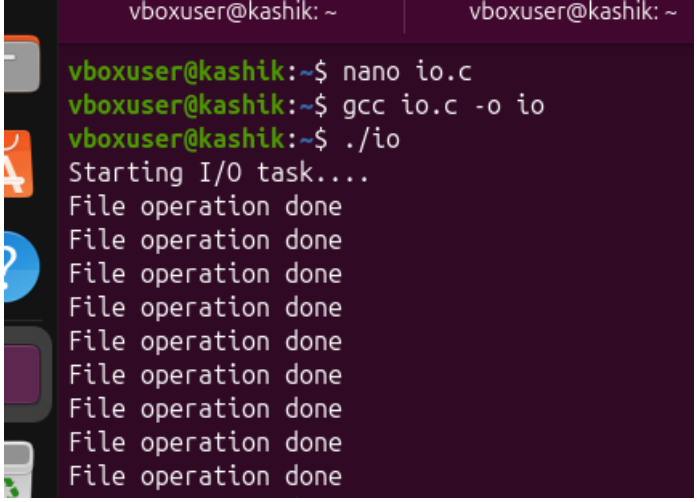
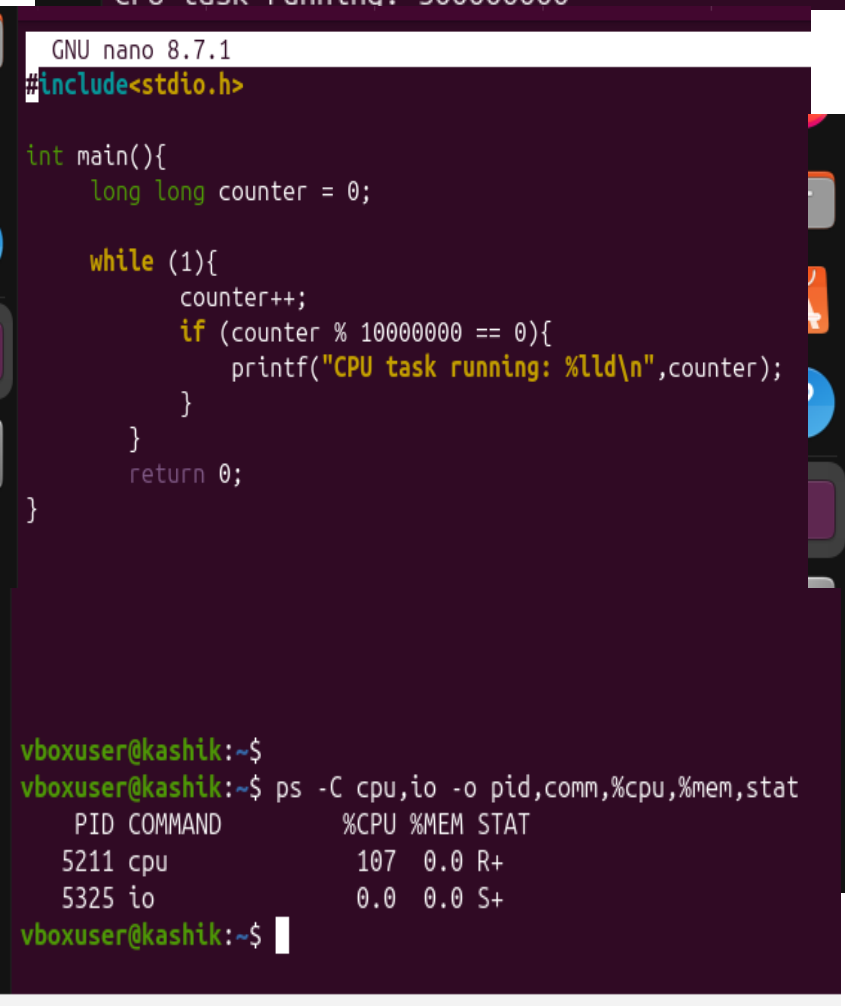
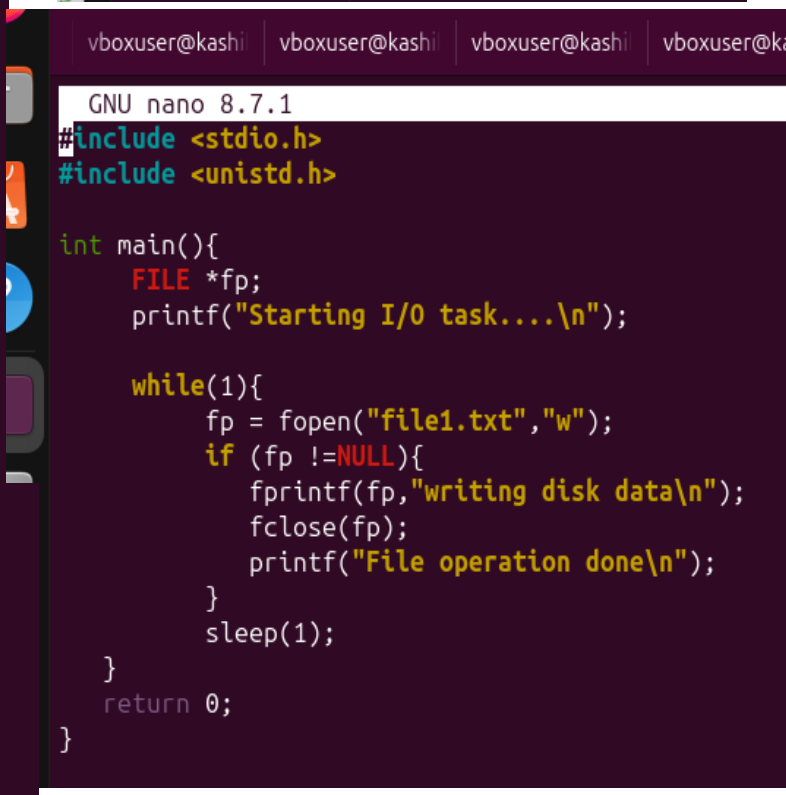
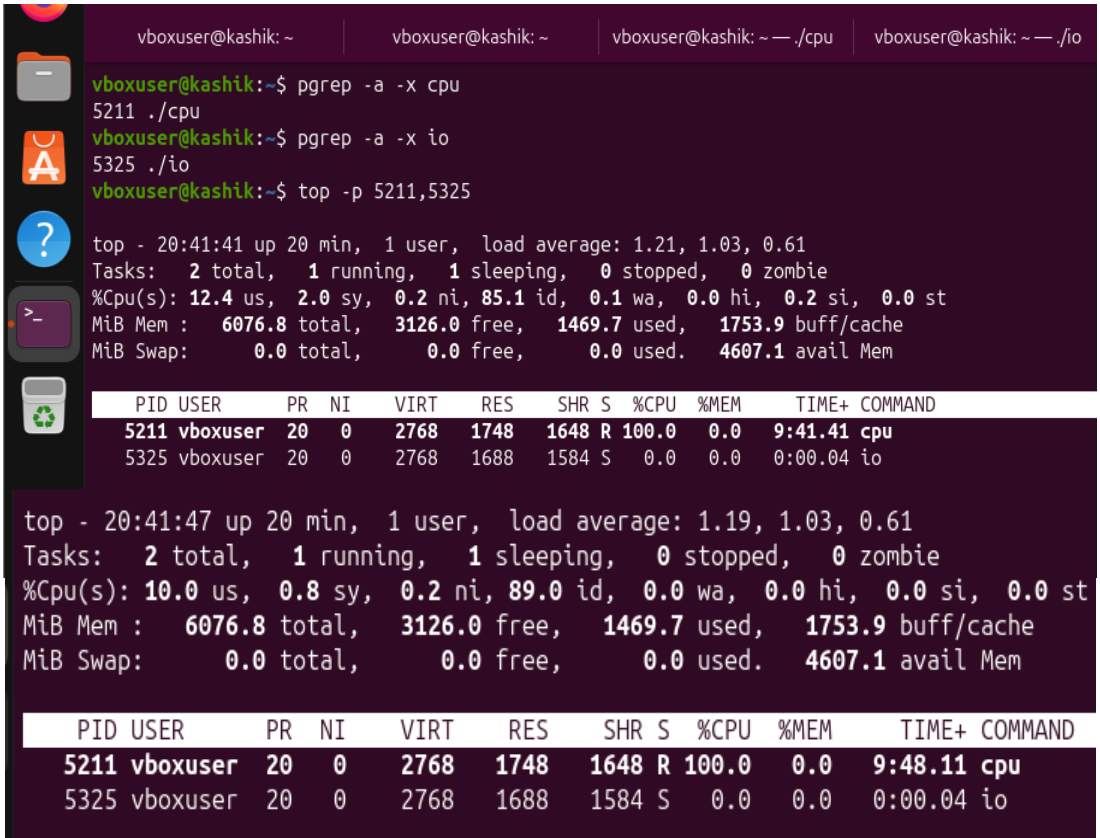

vboxuser@kashik: ~
vboxuser@kashik: ~
vboxuser@kashik:~$ nano cpu.c
vboxuser@kashik:~$ gcc cpu.c -o cpu
vboxuser@kashik:~$ ./cpu
CPU task running: 100000000
CPU task running: 200000000
CPU task running: 300000000
CPU task running: 400000000
CPU task running: 500000000


vboxuser@kashik: ~
vboxuser@kashik: ~
vboxuser@kashik:~$ nano io.c
vboxuser@kashik:~$ gcc io.c -o io
vboxuser@kashik:~$ ./io
Starting I/O task....
File operation done
File operation done
File operation done
File operation done
File operation done
File operation done
File operation done
File operation done
File operation done


GNU nano 8.7.1
#include<stdio.h>
int main(){
long long counter = 0;
while (1){
counter++;
if (counter % 10000000 == 0){
printf("CPU task running: %lld\n",counter);
}
}
return 0;
}
vboxuser@kashik:~$
vboxuser@kashik:~$ ps -C cpu,io -o pid,comm,%cpu,%mem,stat
PID COMMAND %CPU %MEM STAT
5211 cpu 107 0.0 R+
5325 io 0.0 0.0 S+
vboxuser@kashik:~$


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GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main(){
FILE *fp;
printf("Starting I/O task....\n");
while(1){
fp = fopen("file1.txt","w");
if (fp !=NULL){
fprintf(fp,"writing disk data\n");
fclose(fp);
printf("File operation done\n");
}
sleep(1);
}
return 0;
}


vboxuser@kashik: ~
vboxuser@kashik: ~
vboxuser@kashik: ~ — ./cpu
vboxuser@kashik: ~ — ./io
vboxuser@kashik:~$ pgrep -a -x cpu
5211 ./cpu
vboxuser@kashik:~$ pgrep -a -x io
5325 ./io
vboxuser@kashik:~$ top -p 5211,5325
top - 20:41:41 up 20 min, 1 user, load average: 1.21, 1.03, 0.61
Tasks: 2 total, 1 running, 1 sleeping, 0 stopped, 0 zombie
%Cpu(s): 12.4 us, 2.0 sy, 0.2 ni, 85.1 id, 0.1 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 6076.8 total, 3126.0 free, 1469.7 used, 1753.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 4607.1 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
5211 vboxuser 20 0 2768 1748 1648 R 100.0 0.0 9:41.41 cpu
5325 vboxuser 20 0 2768 1688 1584 S 0.0 0.0 0:00.04 io
top - 20:41:47 up 20 min, 1 user, load average: 1.19, 1.03, 0.61
Tasks: 2 total, 1 running, 1 sleeping, 0 stopped, 0 zombie
%Cpu(s): 10.0 us, 0.8 sy, 0.2 ni, 89.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 6076.8 total, 3126.0 free, 1469.7 used, 1753.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 4607.1 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
5211 vboxuser 20 0 2768 1748 1648 R 100.0 0.0 9:48.11 cpu
5325 vboxuser 20 0 2768 1688 1584 S 0.0 0.0 0:00.04 io